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On the Application of Statistical Concepts to the Buffeting Problem*

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(I) Abstract

This study is intended as a step in the direction of an understanding of the problem of buffeting. The buffeting discussed here is of the type encountered when the wake of some part of the airplane interferes with the tail surface, as happens, for example, during the stall. To approach the general buffeting problem immediately does not seem promising; hence, a simpler problem is studied to bring out some general features of the buffeting problem which are believed to be important. The problem studied is the lift force exerted upon a two-dimensional thin airfoil moving in turbulent air. The state of turbulence will later on be restricted to isotropic turbulence, since the characteristics of this type are the best known and simplest. The explicit results gained from the present analysis will not be immediately applicable to tail buffeting, since the model is certainly oversimplified.

The lift exerted upon an airfoil moving in turbulent air is regarded here as the response of a linear system to a random forcing function. The explicit use of the random character of the forces exerted by the turbulent air is believed essential for an understanding and handling of buffeting. This concept obviously extends beyond the applicability of the simple model discussed here in detail. The concepts used in studying the response of a linear system under the action of a random input force are by no means new. For example, the idea of random phases which leads up to the so-called power spectrum goes back to Lord Rayleigh.[1] The theory of generalized harmonic analysis has been developed in connection with communication engineering and related problems, mostly by Wiener[2, 3] and his school. A parallel and often independent development went on in the theory of turbulence, where the same concepts of correlation functions, power spectrum, etc., have been systematically introduced by G. I. Taylor,[4] von Kármán,[5] and others and used extensively since then. In other branches of aeronautics and aeroelasticity, these concepts and their use are comparatively unknown, and this fact excuses the rather lengthy and detailed discussion of simple concepts in the present paper.‡ The author is well aware of the fact that many of the problems discussed and many of the concepts used are not new. It is the main purpose of the paper to call attention to the use of the statistical concepts in aeronautics and to demonstrate their application with a simple example. For a further study of these and similar subjects, as well as for a more complete list of references, the reader is referred to Wiener's work[3] and a paper by S. O. Rice.[6]

(II) The Response of a Mechanical System to a Random Force

To develop the general idea underlying the buffeting analysis, it is most convenient to look first at a simple problem that has been treated before—namely, the motion of a pendulum in a turbulent stream.[7] The same type of analysis covers many more similar problems—for example, the response of an acoustical resonator to random noise, the response of an electrical circuit to electrical noise, Brownian motion of a torsional balance, etc. These problems all have in common the following: The given system is described by a simple differential equation of the oscillation type. That is, let $y(t)$ represent the angle of the pendulum as function of time, the displacement of an oscillating mass, or the current in an electrical circuit. $y(t)$ is determined by the differential equation

$$\frac{d^2y}{dt^2} + \beta\frac{dy}{dt} + \omega_0{}^2 y = F(t) \qquad (1)$$

where β represents a damping parameter, ω_0 is the eigen frequency of the system, and $F(t)$ is an impressed

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‡ Reference should be made to two unpublished M.I.T. theses[12, 13] in which the problem of gust response of an airplane and of atmospheric turbulence was attacked using statistical concepts similar to the ones used in the present study. The author became aware of these theses only recently. For the engineering approach to buffeting see reference 14.

force. If $F(t)$ is given explicitly as a function of time, we deal with a simple inhomogeneous differential equation. In the simplest case let

$$F(t) = Re\{A \cdot e^{i\omega t}\}$$

that is, the force is a periodic function of t. Then the steady-state solution $y(t)$ is found by writing

$$y(t) = Re\{B \cdot e^{i(\omega t + \varphi)}\} \qquad (A, B \text{ real})$$

B and φ are then determined from Eq. (1)

$$\tan \varphi = -\beta\omega/(\omega_0^2 - \omega^2) \tag{2}$$

$$B^2 = \frac{A^2}{(\omega_0^2 - \omega^2)^2 + \beta^2\omega^2} \tag{3}$$

The mean square of $y(t)$ is related to the mean square of $F(t)$ by Eq. (3), since

$$\overline{y^2} = B^2/2; \qquad \overline{F^2} = A^2/2$$

and, hence,

$$\overline{y^2} = \frac{\overline{F^2}}{(\omega_0^2 - \omega^2)^2 + \beta^2\omega^2} = \frac{\overline{F^2}}{|z(\omega)|^2} \tag{4}$$

$|z(\omega)|^2$ is the absolute square of the impedance of the system.

Now assume that $F(t)$ is not a well-behaved sine wave and not periodic, but a random function. That is, $F(t)$ is defined only by certain mean values—for example, $\overline{F^2}$, $\bar{F}$, etc., where

$$\overline{F^n} = \frac{1}{2T}\int_{-T}^{T} F(t)^n \, dt$$

with T sufficiently large. The actual trace $F(t)$ will vary in an irregular fashion, and, hence, it becomes meaningless to ask for the trace $y(t)$. Especially, the phase of $y(t)$ will lose all its meaning. The following question, however, remains: How is the mean square of y related to the mean square of F? In the case of the pendulum in turbulent air the question is: What is the mean square deflection of the pendulum if the mean square velocity fluctuation (the turbulence level) is known?

From Eq. (4) it is evident that, in order to answer this question, we have to know some of the frequency properties of $F(t)$. The necessary information is furnished by the power spectrum $f(\omega)$. $f(\omega)$ is defined in the Appendix; $f(\omega)$ satisfies the relations

$$\overline{F_\omega^2} = \int_0^\omega f(\omega)\, d\omega \quad \therefore \quad \overline{F^2} = \int_0^\infty f(\omega)\, d\omega \tag{5}$$

that is, the first of Eqs. (5) represents the contribution to $\overline{F^2}$ arising from frequencies less than ω. Similarly, we may define a power spectrum of y by

$$\overline{y^2} = \int_0^\infty g(\omega)\, d\omega \tag{6}$$

Generalized Fourier analysis (for example, references 2 and 3) states that, for the case of a function defined by its power spectrum and mean values, Eq. (4) remains valid for the respective power spectra

$$g(\omega) = \frac{f(\omega)}{(\omega_0^2 - \omega^2)^2 + \beta^2\omega^2} = \frac{f(\omega)}{|z(\omega)|^2} \tag{7}$$

Hence, from Eq. (6)

$$\overline{y^2} = \int_0^\infty \frac{f(\omega)\, d\omega}{|z(\omega)|^2} \tag{8}$$

Eq. (8) includes Eq. (4) as a special case.

The relation, Eq. (7), between the power spectra of the output $g(\omega)$ and the input $f(\omega)$ is true for any linear system. $z(\omega)$ denotes the impedance of the linear system, which, of course, is not necessarily of such a simple form as Eq. (4) or (7).

Consequently, in order to obtain the response of a system to the random forcing function, one has to know the impedance of the system and the power spectrum of the forcing function. Then the mean square of the response is obtained from Eq. (8). A proof of this relation is included in the Appendix under heading (6).

If $f(\omega)$ is a slowly varying function compared to the variation of the impedance with ω, Eqs. (7) and (8) can be written approximately as

$$\overline{y^2} \doteq f(\omega_0)\int_0^\infty \frac{d\omega}{(\omega^2 - \omega_0^2)^2 + \beta^2\omega^2} \doteq \frac{\pi}{2}\frac{f(\omega_0)}{\beta\omega_0^2} \tag{9}$$

Hence, the mean square response depends upon the value of the power spectrum of the force at ω_0, the eigen-frequency of the oscillator, and the damping characteristic of the system. Eq. (9) can be thought of as defining $f(\omega)$ by a measurement with a wave analyzer. It is a known and useful equation, for example, in the theory of Brownian motion.

(III) Motion of a Two-Dimensional Airfoil in Incompressible Turbulent Flow

Consider a flat thin airfoil of chord c moving with a constant velocity U through turbulent air. Let x lie along the chord, y along the span of the wing, and z normal to span and chord. The components of the fluctuating turbulent velocity u, v, w are assumed small compared to U. Because of these turbulent fluctuations, a time-dependent apparent angle of attack α exists at the airfoil, and, hence, fluctuating lift forces are produced. The fluctuating angle of attack α is given by

$$\alpha = w/U \tag{10}$$

as long as the fluctuations are small. $\alpha(t)$ now plays the role of the forcing function. The "response" is the fluctuating lift of the airfoil or, better, the lift coefficient $C_L(t)$

In an approximate way the considerations of Section (II) can be carried over to the present problem. To do this it is first necessary to define an "impedance" of the airfoil. Sears[8] has studied the motion of an airfoil to a sinusoidal gust, and his results can be used for our purpose. Sears considers the fluctuation of the velocity of the form

$$w(x\ ,t)/U = \alpha_0 \cdot e^{i\omega[t-(x/U)]}$$

that is, the vertical velocity consists of harmonic waves of angular frequency ω and wave length $2\pi U/\omega$—i.e., a fixed pattern of sinusoidal gusts is passing the airfoil at a velocity U, the flight velocity of the wing. For this velocity distribution Sears finds the following expression for the lift coefficient:

$$C_L(t) = 2\pi\alpha_0 \cdot e^{i\omega t}\varphi(k) \tag{11}$$

where $k = \omega c/2U$ is essentially the ratio of chord length to wave length of the gust. Thus, the function $\varphi(k)$ has the nature of an admittance, relating the forcing function $\alpha(t)$ to the response $C_L(t)$.

Turbulent fluctuations are essentially three-dimensional—that is, u, v, w will be functions of x, y, z, t. For a complete analysis one would therefore require a three-dimensional formula equivalent to Eq. (8) and a result of three-dimensional airfoil theory equivalent to Eq. (11). For the present it seems sufficient to consider only the component w and the dependence upon x, t. Thus, in turbulent flow we consider a fluctuating velocity or angle of attack of the form

$$\alpha(x,\, t) = w(x,\, t)/U$$

If it is now assumed that the turbulence pattern does not change appreciably during a time of the order c/U, then the turbulent angle of attack will also depend upon $t - (x/U)$ only and Sears' results can be applied. This assumption is frequently made in turbulence analysis and requires essentially the condition that the rate of change of the velocity following the particle is small compared with the rate of change of the velocity at a fixed location.

With this assumption we can carry over the analysis of Section (II) directly. If the power spectrum of $\alpha(t)$ is denoted by $f(\omega)$, we can thus write, for the mean square lift coefficient,

$$\overline{C_L{}^2} = 4\pi^2 \int_0^\infty f(\omega)\left|\varphi(k)\right|^2 d\omega$$

The function $\varphi(k)$ is rather complicated

$$\varphi(k) = \frac{J_0(k)K_1(ik) + iJ_1(k)K_0i(k)}{K_1(ik) + K_0(ik)} \tag{12}$$

J_ν and K_ν are Bessel functions.

For the present analysis we require $|\varphi|^2$. To obtain $|\varphi|^2$ from Eq. (12) is fairly involved. The resulting expression is complicated, and it seems more appropriate for the present study to find a simple approximation to $|\varphi|^2$. For this purpose $|\varphi|^2$ was plotted from Sears' paper as a function of k. Fig. 1 demonstrates that $|\varphi|^2$ is well approximated by

$$|\varphi|^2 = 1/(1 + 2\pi k) \tag{13}$$

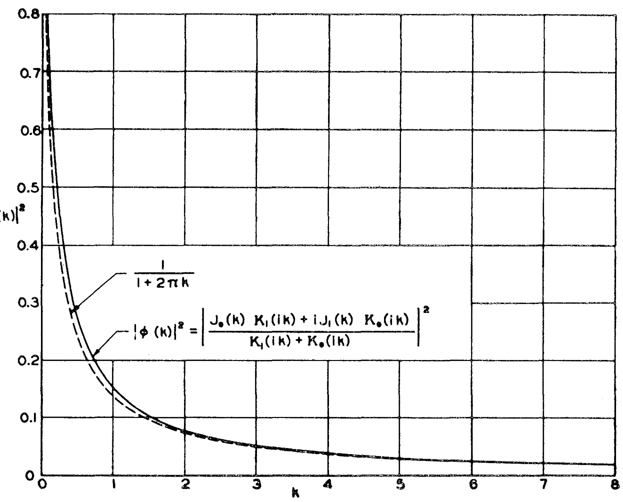

FIG. 1.

Eq. (13) results as an interpolation formula: $|\varphi(0)|^2 = 1$ and using the asymptotic expression in Eq. (12) $|\varphi(k)|^2 \rightarrow 1/2\pi k$ for large k. Using this function the mean square lift coefficient can be written

$$\overline{C_L{}^2} = 4\pi^2 \int_0^\infty \frac{f(\omega)\, d\omega}{1 + (\pi c/U)\omega} \tag{14}$$

$f(\omega)$ denotes the power spectrum of the cross component of turbulence—that is,

$$\overline{\alpha^2} = \frac{\overline{\omega^2}}{U^2} = \int_0^\infty f(\omega)\, d\omega$$

Now for isotropic turbulence it has been found (for example, references 9 and 10) that the bulk of the turbulent energy is contained in a spectrum region that can be described by the simple power spectrum

$$f(\omega) = \frac{\overline{w^2}}{U^2}\frac{L}{\pi U}\frac{1 + 3(L^2\omega^2/U^2)}{[1 + (L^2\omega^2/U^2)]^2} \tag{15}$$

where L denotes the so-called "scale of turbulence" [see the Appendix under heading (2) for definition]. Call $L\omega/U = \xi$; then,

$$f(\omega) = f(0)[(1 + 3\xi^2)/(1 + \xi^2)^2]$$

Using Eq. (15), Eq. (14) can be integrated explicitly. The result is

$$\overline{C_L{}^2} = 4\pi^2\frac{\overline{w^2}}{U^2}\left[\frac{4\eta - \pi}{2\pi(\eta^2 + 1)} + \frac{\eta^2 + 3}{2\pi(\eta^2 + 1)}(\eta \log \eta^2 + \pi)\right] \tag{16}$$

with $\pi c/L \equiv \eta$. Eq. (16) gives the mean square lift coefficient of a two-dimensional airfoil of chord c in isotropic turbulence characterized by the intensity $\overline{w^2}/U^2$ and the scale L* (Fig. 2).

* Actually, one also has to take into account the fact that the turbulence is not two-dimensional. That is, there also exists an effect due to the fact that different parts of the airfoil along the span will find themselves in different phases, and, hence, the overall result is less than Eq. (16). The effect is similar to the length correction of a hot-wire anemometer.

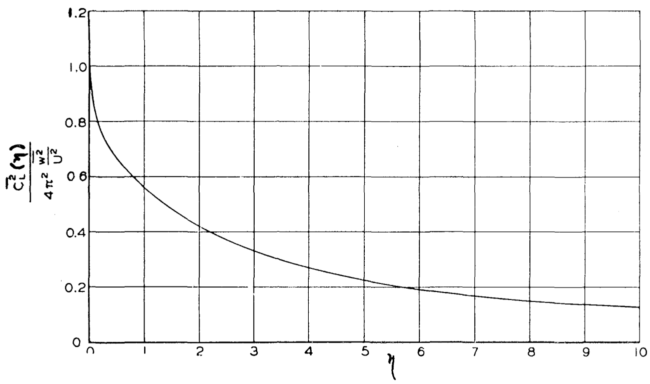

FIG. 2.

Evidently, if $c/L \rightarrow 0$, we have an airfoil of small chord in large-scale turbulence and

$$\overline{C_L^2} \rightarrow 4\pi^2 \frac{\overline{w^2}}{U^2} = 4\pi^2\overline{\alpha^2}$$

The airfoil behaves quasi-stationary. If, on the other hand, c/L becomes large, we deal with an airfoil of long chord in small-scale turbulence. It follows from Eq. (16) that $\overline{C_L^2} \rightarrow 0$. That is to say the "gusts" cancel each other completely and the net lift is zero as might be expected.

A phenomenon not included in an analysis using a spectrum of the form (15) but possibly of major importance for buffeting is the so-called "intermittency" in wake flow.* That is, the edge of any wake fluctuates with a large-scale motion so that a point situated near the edge will sometimes be within the wake and sometimes outside the wake. For a tail surface situated near the edge of the wake of a stalled or partially stalled wing, this "intermittency" may thus be extremely important in determining the lift forces. A detailed analysis would require much more experimental study than is available at present. For a crude idea of the effect, consider the flow at the tail as a region of uniform downwash switched on and off at irregular intervals. Such a flow is probably a good model for the conditions in the wake of an intermittently stalling wing. If the probability of switching over from one régime to the other is assumed to be governed by a Poisson distribution, one has again a simple form of the power spectrum

$$f(\omega) = \frac{\overline{w^2}}{U^2}\frac{2L}{\pi U}\frac{1}{1 + (\omega L/U)^2} \tag{15a}$$

where w denotes the downwash velocity and L/U is the average length of time the downwash is "switched on." The formula corresponding to Eq. (16) is then

$$\overline{C_L^2} = 4\pi^2\frac{\overline{w^2}}{U^2}\frac{2}{\pi}\frac{\eta\log\eta + (\pi/2)}{1 + \eta^2} \tag{16a}$$

* For the concept of "intermittency" see, for example, Townsend, A. A., *The Eddy Viscosity in Turbulent Shear Flow*, Phil. Mag., Vol. 41, p. 320, 1950.

(IV) Aeroelastic Effects

If elastic effects are to be considered, one may ask for the deflection of the wing under the lift forces. Again an "impedance" due to the elastic structure of the wing will enter. Assume, for example, that the elastic character is of the simple type as expressed in Eq. (1). The impedance corresponding to this type of structure has the form

$$(\omega^2 - \omega_0^2)^2 + \beta^2\omega^2 \tag{17}$$

and, hence, the mean square deflection $\overline{y^2}$, due to the fluctuating lift, will be obtained from

$$\overline{y^2} = 4\pi^2\int_0^\infty \frac{f(\omega)\,d\omega}{[1 + \pi(c/U)\omega][(\omega^2 - \omega_0^2)^2 + \beta^2\omega^2]} \tag{18}$$

In more general terms we have

$$\overline{y^2} = \int_0^\infty f(\omega)\psi(\omega)\chi(\omega)\,d\omega \tag{19}$$

where $f(\omega)$ denotes the power spectrum of the turbulence, $\psi(\omega)$ is the absolute square of the admittance of the wing, and $\chi(\omega)$ is absolute square of the admittance of the elastic structure.

Structural deflections will, of course, introduce additional aerodynamic effects. For example, oscillation of the fuselage due to the tail loads will introduce plunging motions of the tail surface and thus aerodynamic damping of the oscillation. These coupling and feedback effects will vary from case to case and may sometimes be rather complex.

From Eq. (18), for small damping, follows the simple relation:

$$\overline{y^2} = 4\pi^2\frac{\pi f(\omega_0)}{2\beta\omega_0^2[1 + \pi(\omega_0 c/U)]} \tag{20}$$

For the simple case of the input spectrum (15) this yields

$$\overline{y^2} = \frac{\overline{w^2}}{U^2}\,2\,\frac{\pi L}{U}\frac{1 + 3(L^2\omega_0^2/U^2)}{[1 + \pi(\omega_0 c/U)][1 + (L^2\omega_0^2/U^2)]^2} \tag{21}$$

A result like Eq. (20) applies within the limits of the analysis, for example, to the response of an elastically restrained flat plate used for turbulence measurements.†

(V) Response to a Vortex Street

Sometimes buffeting is assumed to be caused by a von Kármán vortex street. While this may be the case in specific instances, it does not seem to be the general case. Even if a vortex street of nearly regular frequency is present, little will be gained by a detailed analysis involving the phase of the lift, as well as its magnitude. From the general formulation for $\overline{C_L^2}$ given here, the mean square lift due to a vortex street is, of course, obtainable. The velocity components in a regular

† A device similar to this has been used by Ackeret for turbulence measurements in water (verbal communication).

vortex street are sinusoidal, and, hence, they have a power spectrum

$$f(\omega) = (A^2/2)\delta(\omega - \Omega) \quad (\delta \equiv \delta \text{ function}) \quad (22)$$

where Ω represents the vortex frequency. Inserting this into Eq. (14) yields

$$\overline{C_L^2} = 2\pi A^2\{1/[1 + (\pi c/U)\Omega]\} \quad (23)$$

This result could have been obtained directly from Sears' analysis.

In most cases even a vortex street shows a continuous background spectrum. This is clearly seen in Fig. 3, where the power spectrum as measured behind a cylinder at a Reynolds Number of 4×10^3 is shown. Note the distribution of energy between the continuous part and the discrete band. The spectrum shown is the one corresponding to the fluctuation in the direction of the mean flow. Hence, the simple analytical spectrum corresponding to Eq. (15) is simply of the form

$$f(\omega) = a/(1 + b\omega^2)$$

[compare with the Appendix under heading (5)]. The relative energy in the shedding frequency band and the continuous part is proportional to the relative area as indicated on Fig. 3.

The continuous part of the spectrum obviously becomes more predominant further downstream from the cylinder.*

(VI) The Probability and Frequency of Excessive Values of the Response

It will often become necessary to ask not only for the mean square response but also for the probability that a certain value of y is exceeded. This question is natural for strength considerations. If the probability distribution of the response is known, then this answer can be given immediately. Often it will be possible to assume, at least approximately, a Gaussian distribution of the response. In this case the probability that the variable y will exceed k times the root mean square value $\sqrt{\overline{y^2}}$, for large k, is

$$\text{prob.}\left\{y > k\sqrt{\overline{y^2}}\right\} \cong \frac{e^{-(1/2)(k^2)}}{k\sqrt{2\pi}} \quad (24)$$

For example, the probability to exceed $3\sqrt{\overline{y^2}}$ is about 0.002. One can also give some general, but also broad, limits for the probability that a certain value is exceeded without knowing the distribution explicitly. For example, according to Chebycheff's inequality,[11] one has

$$\text{prob.}\left\{|y - \bar{y}| > k\sqrt{\overline{y^2}}\right\} \leq 1/k^2 \quad (25)$$

* Fig. 3 was taken from a paper "On the Development of a Turbulent Wake from a Vortex Street," by A. Roshko, prepared under N.A.C.A. Contract NAw-6048 at the Guggenheim Aeronautical Laboratory, California Institute of Technology. This paper will appear soon as an N.A.C.A. publication.

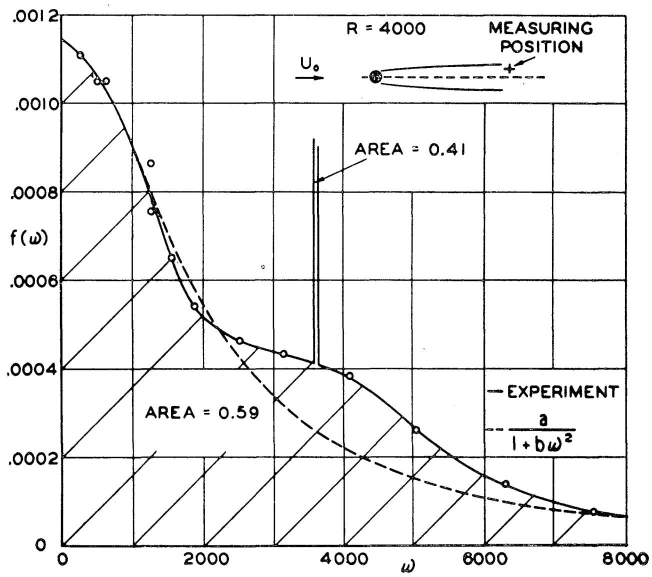

Fig. 3.

The probability that y differs from its mean value by k times the root mean square is less or equal to $1/k^2$. Applying this to the lift coefficient for the wing as computed here yields

$$\text{prob.}\left\{C_L \geq k\sqrt{\overline{C_L^2}}\right\} \leq 1/k^2$$

Thus, C_L will be ten times larger than $\sqrt{\overline{C_L^2}}$ in less than one out of a hundred cases. The Chebycheff inequality is known to be far too broad for most practical applications, and the upper limit given is, in general, much too high. The estimate can be considerably improved if one knows more about the distribution function. In the present application, for example, it is safe to use Gauss' estimate with zero skewness,[13] which gives

$$\text{prob.}\left\{C_L \geq k\sqrt{\overline{C_L^2}}\right\} \leq 4/9k^2 \quad (26)$$

Further improvements on this estimate are possible.[11]

For fatigue considerations it will also be useful to ask for the number of times per unit time a certain value is exceeded. In a number of cases the answer to this question can be given explicitly. The average number of "ξ values" N_ξ of a stochastic function $I(t)$—that is, the number of times $I(t)$ passes the value ξ—is given by [6]

$$N_\xi = \int_{-\infty}^{+\infty} \rho(\xi, \eta)|\eta|\, d\eta \quad (27)$$

where $\rho(\xi, \eta)\, d\xi\, d\eta$ denotes the probability of finding $I(t)$ between ξ and $\xi + d\xi$ with a derivative $I'(t)$ between η and $\eta + d\eta$. If $I(t), I'(t)$ are Gaussian, N_ξ becomes simply

$$N_\xi = (1/\pi)e^{-\xi^2/2\overline{\xi^2}}\sqrt{\overline{\eta^2}/\overline{\xi^2}}$$

The average number of times the function exceeds $k\sqrt{\overline{\xi^2}}$ is thus given by

$$\frac{1}{2} N_{k\sqrt{\overline{\xi^2}}} = \frac{1}{2\pi} e^{-k^2/2} \sqrt{\frac{\overline{\eta^2}}{\overline{\xi^2}}} \tag{28}$$

The factor $^1/_2$ stems, evidently, from the fact that $I(t)$ passes the value ξ half of the time with positive slope and half of the time with negative slope. Only the former enters here. In terms of the power spectrum $g(\omega)$ of I, the result is

$$\frac{1}{2} N_{k\sqrt{\overline{\xi^2}}} = \frac{e^{-k^2/2}}{2\pi}\left[\int_0^\infty \omega^2 g(\omega)\, d\omega \Big/ \int_0^\infty g(\omega)\, d\omega\right]^{1/2} \tag{29}$$

(VII) Concluding Remarks

It is believed that, in spite of the simplicity of the model discussed here, a few important features of the buffeting problem have been brought out as follows:

(1) Since one deals with a phenomenon that is largely random in character, there is no use in making an analysis that will give the phase of the motion. Only mean quantities are defined and useful.

(2) The use of the power spectrum concept for the motion of the air and of the impedance concepts for the aerodynamic and elastic response makes the analysis relatively simple.

For a complete theory of buffeting, obviously much more analysis is required and, especially, considerably more knowledge of the turbulence structure of the flow in wakes. The latter is very much an experimental program. However, it is likely that, once there are some data obtained on the flow in the wake of a stalled wing, for example, one may find simple rules for the power spectrum and turbulent intensities. For example, the turbulent intensity must be related to the drag of the body that produces the wake; the scale, to its lateral dimensions.

It should be pointed out that there exists a number of other problems in aeronautics for which the concepts used here are the natural ones. For example, wave drag of rough surfaces and, in general, the definition and measurement of aerodynamical roughness belong here. The motion of smoke stacks and steel towers, as well as airplanes in gusty air, also belongs to this type of phenomenon. (The gust problem for aircraft has recently been attacked using the technique of stationary random processes.[12, 13]) Taxiing over rough ground is another problem in the same class.

Appendix

(1) Mean Values

Assume that velocity fluctuations in turbulent flow are observed with a hot-wire anemometer and are reproduced on the screen of an oscilloscope. Let $u(t)$ be the fluctuation measured at a particular point in the flow. $u(t)$, for example, as viewed on the screen of the oscilloscope, will show an irregular character, and no periodicities will be apparent. If an instantaneous picture of the screen is taken repeatedly in intervals, one will find that the trace $u(t)$ will differ from picture to picture. Hence, it does not make sense to ask for the complete function $u(t)$; only certain *mean values* are reproducible and measurable quantities. One may find the mean values of $u(t)$ in the following way: Take a large number, say N, pictures of the screen and note the value of $u(t)$ at the center of each picture. This gives N values for u_i. The mean value $\bar{u}$ is given by

$$\bar{u}(t) = (u_1 + u_2 + \ldots u_N)/N$$

To be more precise one should define $\bar{u}(t)$ as the limiting value for $N \to \infty$, that is

$$\bar{u}(t) = \lim_{N\to\infty} \left(\sum_0^N u_i/N\right)$$

For turbulent flow $u(t)$ will be negative as often as positive and $\bar{u}(t) = 0$. However, the mean value of $u^2(t)$ will, in general, be finite and represents a measure for the *intensity* of turbulence or at least of the particular component of turbulence measured. Again, $\overline{u^2}(t)$ is defined by

$$\overline{u^2}(t) = \lim_{N\to\infty} \left(\sum u_i^2/N\right)$$

Now, if turbulence is observed at a fixed position in a steady mean airflow, then the fluctuating field will be *stationary* in the sense that the mean values do not depend upon the time—that is, $\overline{u^2}$ at a specific point in the field will have the same value at any time. Clearly, many flows of interest have this property; the flow in the wake of a stalled airfoil obviously has this property if the stall persists for a large length of time compared with the characteristic times of the turbulence in the wake. This will certainly be true for a slow stall and even in many cases of pull-out.

For the analysis in this paper it is always assumed that the fluctuations are *stationary*. It can be shown that in this case the mean values defined above in the way of *ensemble averages* are identical with the *average taken over time* provided the time scale T [see (2) below] is finite, and similar integrability conditions apply to higher order correlations.

(2) Correlation Coefficient

Assume again that N pictures of the turbulent trace $u(t)$ are taken. Take the value of u at a particular point on each picture (for example, the center) and call it $u(t)$. Then take the value of u at a neighboring point $t + \tau$, say, and call this $u(t + \tau)$. Form the average of the product of the two—that is,

$$\overline{u(t)\,u(t+\tau)} = \lim_{N\to\infty} [\sum_0^N u_i(t)\,u_i(t+\tau)/N]$$

If the process is *stationary*, then the product will again be independent of t and a function of τ only. This means that the mean product will depend only upon the distance between the points chosen on each picture but not *where* they are chosen. Hence, we find a function

$$\psi(\tau) = \lim_{N\to\infty} [\sum_0^N u_i(t)\,u_i(t+\tau)/N]$$

$\psi(\tau)$ is called the *correlation function* or, more exactly, *autocorrelation function*.

It is easy to establish some properties of ψ. If $\tau = 0$, we have

$$\psi(0) = \lim_{N\to\infty} \frac{\sum_0^N u_i(t)\,u_i(t+\tau)}{N} = \lim_{N\to\infty} \frac{\sum_0^N u_i^2(t)}{N} = \overline{u^2}$$

If τ becomes large, we will often expect $\psi \to 0$, since, if the process under consideration is random in character, we do not expect any correlation between values of $u(t)$ at t and, say, of $u(t)$ a half an hour later. Hence, if τ is large, it is equally likely that a positive value of $u(t)$ will be multiplied by a positive or negative value $u(t+\tau)$, and, hence, the sum will come out to zero. Hence, we expect $\psi(\tau)$ to have a behavior as shown in Diagram A. Whether the function approaches zero monotonically or oscillating depends upon the specific process. Furthermore ψ must be an even function—that is, $\psi(\tau) = \psi(-\tau)$. Clearly, $\psi(\tau)$ is some measure of the "scale" of the process—that is, the time distance over which there exists a dependency between mean values of the fluctuation. In this sense ψ has a meaning for the definition of "eddy size," etc. To define the "scale," or better "time scale," of the motion more precisely, one defines a time T by the *area* under the correlation curve and writes

$$\psi(0)T = \int_0^\infty \psi(\tau)\,d\tau$$

(3) ***Space and Time Correlations***

So far, the correlation has been defined in time only. Clearly, in homogeneous turbulence one may also define stationary correlations in space. Instead of choosing t and $t + \tau$, one may measure $u(t)$ at x, y, z and $x + \Delta x$, $y + \Delta y$, $z + \Delta z$ and form a correlation function; or one may pick a different velocity component $v(t)$ or $w(t)$, and, hence, *several different* correlation functions may be defined. It is here that the simplification due to an isotropic turbulent field enters. Here, as shown first by Taylor and by von Kármán, two correlation functions are sufficient to give all of the others. The two chosen are ordinarily denoted by f and g and are defined in easily understood picture language by Diagram B—that is, $f(r)$ is the correlation between

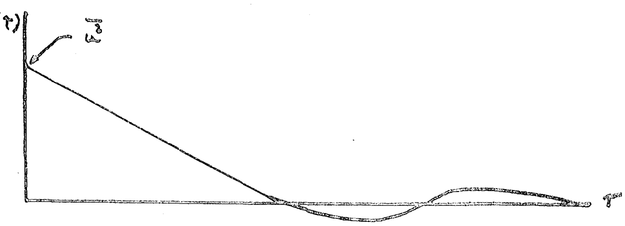

DIAGRAM A.

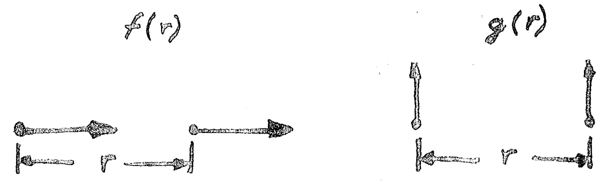

DIAGRAM B.

velocity components *along the line in the direction* of the component. $g(r)$ is the correlation between points chosen at a distance r in the direction *normal* to the direction of the velocity components. We have

$$\overline{u(x, y, z, t)\quad u(x + r, y, z, t)} = f(r)$$
$$\overline{u(x, y, z, t)\quad u(x, y + r, z, t)} = g(r)$$
$$\overline{v(x, y, z, t)\quad v(x + r, y, z, t)} = g(r)$$

If the field of fluctuation u, v, w is superimposed upon a mean flow of velocity U in the x-direction and, furthermore, if $\overline{u^2}/U^2$, $\overline{v^2}/U^2$, $\overline{w^2}/U^2$ are small quantities, then it is often possible to interchange time and space variables and consider the field of turbulence simply transported along with the velocity U along x. Consequently, the time τ entering into the correlation function can be replaced by r/U, where r is chosen along the x-axis. If this is done, there results a unique relation between $\psi(\tau)$ and $g(r)$ and $f(r)$, respectively. For example,

$$\overline{u(t)\quad u(t+\tau)} = \psi_1(\tau) = f(r/U)$$
$$\overline{v(t)\quad v(t+\tau)} = \psi_2(\tau) = g(r/U)$$

(4) ***Concept of Power Spectrum***

Assume that the time correlation function $\psi(\tau)$ is known. Form the cosine transform $f(\omega)$ of $\psi(\tau)$,

$$f(\omega) = \frac{2}{\pi}\int_0^\infty \psi(\tau)\cos\omega\tau\,d\tau$$

This function $f(\omega)$ is identical with the power spectrum introduced before. This becomes plausible if one observes a few properties of $\psi(\tau)$ as defined above. For example, using the inversion formula for a cosine transform, we have

$$\psi(\tau) = \int_0^\infty f(\omega)\cos\omega\tau\,d\omega$$

and, hence,

$$\psi(0) = \int_0^\infty f(\omega)\, d\omega$$

Since

$$\psi(0) = \overline{u^2}$$

we have

$$\overline{u^2} = \int_0^\infty f(\omega)\, d\omega$$

(5) *Simple Forms of the Power Spectrum*

Experiments[9, 10] have shown that in isotropic turbulence the correlation functions $f(r)$ and $g(r)$ have nearly the form

$$\left.\begin{aligned} f(r) &= f(0)e^{-r/L} \\ g(r) &= g(0)e^{-r/L}\,[1 - (r/2L)] \end{aligned}\right\} r > 0$$

Hence, the corresponding time correlation functions are

$$\psi_1(\tau) = \psi_1(0)e^{-\tau U/L}$$

$$\psi_2(\tau) = \psi_2(0)e^{-\tau U/L}\,[1 - (U/2L)\tau]$$

The corresponding spectra are

$$\left.\begin{aligned} f_1(\omega) &= f_1(0)\,\frac{1}{1+\xi^2} \\ f_2(\omega) &= f_2(0)\,\frac{1+3\xi^2}{(1+\xi^2)^2} \end{aligned}\right\} \xi = \frac{\omega L}{U}$$

In the analysis we dealt with the angle of attack distribution in the x-direction [Section (III)]. In this case we want

$$\overline{w(x, t) \quad w(x + r, t)}$$

that is, $g(r)$ and, hence, $\psi_2(\tau)$ or $f_2(\omega)$.

(6) *Proof of Eq.* (8)

For completeness, a short proof of Eq. (8) will be included here. It is convenient here to write the Fourier integrals in the complex form. Let

$$\mathcal{L}(y) = F(t)$$

denote a linear differential equation relating $y(t)$ to a forcing function $F(t)$. Denote by $f(\omega)$, $\varphi(\tau)$, the power spectrum and autocorrelation function of $F(t)$; and by $g(\omega)$, $\psi(\tau)$, the power spectrum and correlation function of $y(t)$. We have

$$\overline{F^2} = \int_0^\infty f(\omega)\, d\omega = \varphi(0)$$

$$\overline{y^2} = \int_0^\infty g(\omega)\, d\omega = \psi(0)$$

$$\varphi(\tau) = \frac{1}{2}\int_{-\infty}^{+\infty} f(\omega)\cdot e^{i\omega\tau}\, d\omega$$

$$\psi(\tau) = \frac{1}{2}\int_{-\infty}^{+\infty} g(\omega)\cdot e^{i\omega\tau} d\omega$$

Call $h(t)$ the response of the linear system to a unit impulse at $t = 0$. For a process going on since $t = -\infty$ and for a system with positive damping, the solution $y(t)$ can be written

$$y(t) = \int_{-\infty}^{t} F(\tau)\, h(t - \tau)\, d\tau$$

Since $h(t)$ is zero for $t < 0$, the upper limit of the integral can be extended to $+\infty$. The variable $t - \tau$ can be changed to σ, say. Thus,

$$y(t) = \int_{-\infty}^{+\infty} F(\tau)h\,(t - \tau)\, d\tau = \int_{-\infty}^{+\infty} F(t - \sigma)\, h(\sigma)\, d\sigma$$

The correlation function

$$\psi(\xi) = \overline{y(t) \;\; y(t + \xi)}$$

using the above integral becomes

$$\overline{y(t)\, y(t + \xi)} = \iint_{-\infty}^{+\infty} \overline{F(t - \sigma)\, F(t + \xi - \sigma')}\, h(\sigma)\, h(\sigma')\, d\sigma\, d\sigma'$$

or

$$\psi(\xi) = \iint_{-\infty}^{+\infty} \varphi(\xi + \sigma - \sigma')\, h(\sigma)\, h(\sigma')\, d\sigma\, d\sigma'$$

Replacing ψ and φ by the Fourier integral involving the respective power spectra yields

$$\int_{-\infty}^{+\infty} g(\omega)e^{i\omega\xi}\, d\xi = \iiint_{-\infty}^{+\infty} f(\omega)e^{i\omega\xi}\, h(\sigma)\cdot e^{i\omega\sigma}\cdot h(\sigma')e^{-i\omega\sigma'}\, d\sigma\, d\sigma'$$

Now the impedance $Z(\omega)$ is defined as the ratio of input over output for a sinusoidal input. Hence,

$$\frac{1}{z(\omega)} = \int_{-\infty}^{+\infty} h(\sigma)e^{i\omega\sigma}\, d\sigma$$

and thus

$$\int_{-\infty}^{+\infty} g(\omega)e^{i\omega\xi} = \int_{-\infty}^{+\infty} f(\omega)e^{i\omega\xi}\, \frac{1}{z(\omega)}\, \frac{1}{z(\omega)^*}\, d\omega$$

$$g(\omega) = f(\omega)/|z|^2$$

References

[1] Lord Rayleigh, *Theory of Sound*, p. 371; Dover Publications, New York, 1945.

[2] Wiener, N., *The Fourier Integral*, p. 150; Cambridge, 1939.

[3] Wiener, N., *Extrapolation, Interpolation and Smoothing of Stationary Time Series;* John Wiley & Sons, Inc., New York, 1950.

[4] Taylor, G. I., *Statistical Theory of Turbulence*, Proc. Roy. Soc., Series A, Vol. 151, p. 421, 1935.

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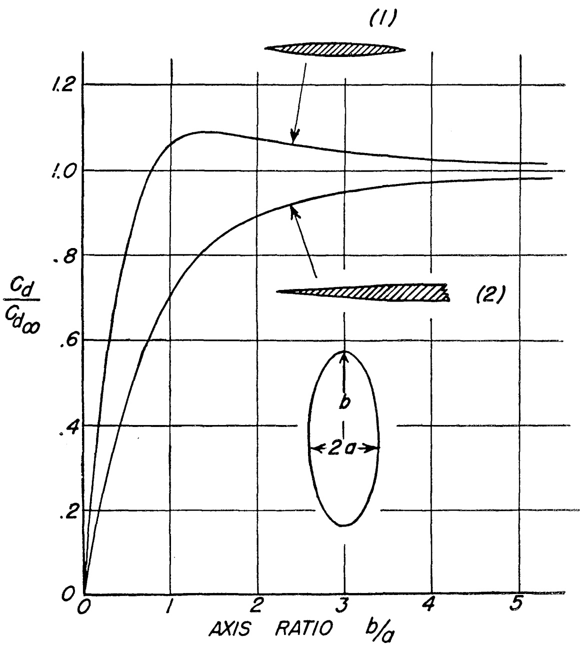

FIG. 8. Minimum drag of elliptic wings having (1) a given volume and (2) a given base area.

$$C_{D_{min.}} = \frac{t_0^2}{a^2}\left(\frac{1}{\sqrt{1 + (a^2/b^2)}}\right)\left(1 + 2\frac{a^2}{b^2}\right) \quad (71)$$

In this case the sections have a constant curvature in the stream direction. Eqs. (70) and (71) are plotted in Fig. 8.

In each case the area of the cross sections has the same distribution along x as that of the corresponding optimum body of revolution. It is interesting to note that each of these figures yields the minimum drag for all distributions of thickness within the space defined by the intersection of its forward and reversed characteristic envelopes.

REFERENCES

[1] Munk, Max. M., *The Minimum Induced Drag of Airfoils*, N.A.C.A. T.R. No. 121, 1921.

[2] Jones, Robert T., *The Minimum Drag of Thin Wings in Frictionless Flow*, Journal of the Aeronautical Sciences, Vol. 18, No. 2, p. 75, February, 1951.

[3] Whittaker, E. T., and Watson, G. N., *Modern Analysis*, pp. 388–403; The Macmillan Company, New York, 1945.

[4] Bergman, Stefan, *Classes of Solutions of Linear Partial Differential Equations in Three Variables*, Duke Mathematical Journal, Vol. 13, No. 3, September, 1946.

[5] Jones, Robert T., *Leading Edge Singularities in Thin Airfoil Theory*, Journal of the Aeronautical Sciences, Vol. 17, No. 5, pp. 307–310, May, 1950.

[6] von Kármán, Th., *Supersonic Aerodynamics—Principles and Applications*, Journal of the Aeronautical Sciences, Vol. 14, No. 7, pp. 373–409, July, 1947.

[7] Hayes, Wallace D., *Linearized Supersonic Flow*, North American Aviation, Inc., Los Angeles, Report No. A.L. 222, June, 1947.

[8] Bateman, H., *Partial Differential Equations*, pp. 95–100; Dover Publications, New York, 1944.

[9] Bierens de Haan, D., *Nouvelles Tables D'Integrals Definies*, Table 68, pp. 104–105; G. E. Stechert & Company, New York, 1939.

[10] Hobson, E. W., *Spherical and Ellipsoidal Harmonics*, pp. 360–364; Cambridge University Press, 1931.

[11] Busemann, A., *Infinitesimale Kegelige Uberschallstromung*, Yearbook of the German Academy for Aeronautical Research, Vol. 713, No. 3, pp. 105–122, 1943.

On the Application of Statistical Concepts to the Buffeting Problem

(*Concluded from page 800*)

[5] von Kármán, Th., *On the Statistical Theory of Turbulence*, Proc. Nat. Acad., Vol. 23, p. 98, 1937.

[6] Rice, S. O., *Mathematical Analysis of Random Noise*, Bell Technical Journal, Vol. 18, p. 282, 1944, and Vol. 19, p. 46, 1945.

[7] Lin, C. C., *On the Motion of a Pendulum in Turbulent Flow*, Quart. Appl. Math., Vol. 1, p. 43, 1943.

[8] Sears, W. R., *Some Aspects of Nonstationary Airfoil Theory and Its Practical Applications*, Journal of the Aeronautical Sciences, Vol. 8, No. 2, p. 104, February, 1941.

[9] Dryden, H. L., *Turbulence Investigations at the National Bureau of Standards;* Proc., Fifth International Congress of Applied Mechanics, Cambridge, 1938.

[10] Liepmann, H. W., Laufer, J., and Liepmann, K., *On the Spectrum of Isotropic Turbulence*, N.A.C.A. T.N. No. 2473, 1951.

[11] Cramér, H., *Mathematical Methods of Statistics;* Princeton Univ. Press, Princeton, N.J., 1951.

[12] Clementson, G. C., *An Investigation of the Power Spectral Density of Atmospheric Turbulence*, M.I.T. Ph.D. Thesis, 1950.

[13] Clement, W. F., and Chippendale, G. R., *A Statistical Study of Atmospheric Turbulence by Flight Measurements*, M.I.T. Master Thesis, 1951.

[14] Luskin, H., and Lapin, E., *An Analytical Approach to the Fuel Sloshing and Buffeting Problems of Aircraft*, Journal of the Aeronautical Sciences, Vol. 19, No. 4, p. 217, April, 1952.